MIL–L–12997C
27 MARCH 1964

SUPERSEDING
MIL–L–12997B
12 DECEMBER 1958

MILITARY SPECIFICATION

LINK, CARTRIDGE, METALLIC BELT, 20MM, M12

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of metallic belt link (welded or nonwelded) intended to hold 20MM ammunition in the form of flexible belts for use in the M39A2 20MM automatic gun.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL–P–116 —Preservation, Methods of.
MIL–W–13855 —Weapons, Small Arms, General Specification for.
MIL–W–45223 —Welding, Spot, Hardenable Steels.

STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 151 —Metals; Test Methods.

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–109 —Quality Assurance Terms and Definitions.

DRAWINGS

U.S. ARMY WEAPONS COMMAND

D7147027 —Body, Inner.
D7147028 —Link, Cartridge, Metallic Belt, 20MM, M12.
D7548085 —Link, Cartridge, Metallic Belt, 20MM, M12 (Alternate).
F7190358 —Pull Test Fixture Gage.
F7190403 —Test Fixture Gage.
IEL7101289 —Index of Inspection Equipment Lists.

PUBLICATIONS

U.S. ARMY WEAPONS COMMAND

P7147028 —Packaging Data Sheet for Link, Cartridge, Metallic Belt, 20MM, M12.
P7548085 —Packaging Data Sheet for Link, Cartridge, Metallic Belt, 20MM, M12.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1305

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3. REQUIREMENTS

3.1 Production samples. A sample of 800 links representing initial production shall be packaged in accordance with contract requirements and identified with drawing and specification numbers, revision numbers and dates, contract number, name of contractor, and district office. Thereafter, a sample of 400 links representing each month's production shall be packaged level C in accordance with Packaging Data Sheet P7147028 or P7548085, as applicable, and identified as specified above. Monthly samples shall be submitted until three consecutive monthly samples have been determined to be satisfactory by the testing agency. Accompanied by copies of the required certifications of chemical analyses, physical tests, etc., samples shall be forwarded prepaid by the contractor to Springfield Armory, Springfield, Massachusetts 01101, Attn: SWESP–QA. The initial production sample shall be contained in the pilot pack (see 5.1). Sample links shall be representative of the normal production process utilized by the contractor and representative of materiel which has met the requirements, exclusive of functioning (see 3.2.7), and which has been inspected and determined to be acceptable. Sample links will not be returned to the contractor except as evidence to support the testing agency's report.

3.2 Materials, construction, and design. Links shall conform to material, construction, and design requirements specified herein and on Drawing D7147028 or D7548085, as applicable, and drawings referenced thereon, and the materials and construction shall be in accordance with the applicable provisions of Specification MIL–W–13855. Finished links shall be free of cracks, tears, and malformations, as defined in 6.5, when examined as specified in 4.3.2.3.2.

3.2.1 *Heat treatment.* Links shall be heat treated to produce a hardness within the hardness range specified on the applicable drawings. Heat treatment shall be conducted in such a manner as to prevent formation of carburized layers or decarburization in excess of the limits specified on the applicable drawings. Heat treatment processes other than oil quenching and tempering shall not be performed by the contractor without prior approval of the procuring agency.

3.2.2 *Weld (applicable to welded links only).* Resistance weld of the stress bridge shall be capable of meeting the weld test requirement specified on the applicable drawing when tested as specified in 4.3.3.4 and 4.4.2.

3.2.3 *Pull.* The assembled links shall be capable of meeting the pull test requirements specified on Drawing D7147028 or D7548085, as applicable, when tested as specified in 4.3.3.5 and 4.4.3.

3.2.4 *Twist.* Links assembled into belts shall be capable of withstanding the twist test specified in 4.3.3.6 and 4.4.4 without evidence of partial or complete belt separation or disassembly of cartridges from the links.

3.2.5 *Cartridge retention.* Completed links shall meet the cartridge retention requirement specified on the applicable drawing when tested as specified in 4.3.3.7 and 4.4.5.

3.2.6 *Free hinging.* Links assembled into belts shall hinge without binding and fold over smoothly without interference between the projecting surfaces of the links when tested as specified in 4.3.3.8 and 4.4.6.

3.2.7 *Functioning.* Links shall be capable of being assembled into belts and shall operate without malfunctioning, cracking, breaking, or separating when tested as specified in 4.3.3.9 and 4.4.7.

3.3 Marking. Each link shall be clearly marked in accordance with the applicable drawing and Specification MIL–W–13855. The contractor shall furnish the Government representative an accurate record of the data pertaining to the changes in the lot numbers (see 6.3).

3.4 Workmanship. Workmanship shall be in accordance with the workmanship requirements of Specification MIL–W–13855.

4. QUALITY ASSURANCE PROVISIONS

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4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Contractor's inspection system.* The contractor shall provide and maintain an inspection system in accordance with Specification MIL–I–45208.

4.1.2 *Quality assurance terms and definitions.* Quality assurance terms and definitions used herein are in accordance with Standard MIL–STD–109.

4.2 **Production sample inspection.** Inspection of production samples (see 3.1) and pilot pack (see 5.1) shall be performed at the designated testing agency. Production samples and pilot pack shall be subjected to the examination and tests specified herein, including the functioning test (see 4.3.3.9 and 4.4.7), and such other examination and tests as are necessary to determine compliance with the requirements of the contract. Acceptance of the initial lot (see 3.1) shall be contingent upon approval of the initial production samples. Acceptance of subsequent lots shall not be contingent upon approval of the monthly samples. The contractor will be informed of the results of any tests which indicate failure of the links to meet prescribed requirements.

4.3 **Inspection provisions.**

4.3.1 *Inspection lot.* The formation, size, and presentation of inspection lots shall be in accordance with Standard MIL–STD–105. Inspection lots shall be as large as practicable, in consideration of quality history, manufacturing conditions, and contractor's delivery schedule, and within the limitations of Specification MIL–W–13855.

4.3.2 *Examination.* Examination of links and packaging shall be performed in accordance with 4.3.2.3, and the applicable inspection provisions of Specification MIL–W–13855.

4.3.2.1 *Links.* Sample size shall be in accordance with Standard MIL–STD–105, using inspection level II.

4.3.2.2 *Packaging.* Sample size shall be in accordance with Standard MIL–STD–105, using inspection level I.

4.3.2.3 *Classification of defects.* Classification of defects and acceptable quality levels (AQL's) for examination in accordance with Standard MIL–STD–105 shall be as specified herein and shall include the following provisions:

(a) The AQL listed for each defect shall be applied to the individual defect, not to a group of defects.

(b) Listings under "Zone on Drawing" are given only as an aid in locating applicable dimensions or values.

(c) The presence of nicks, burrs, scratches, tool marks, deformations, missing operations, or other signs of poor workmanship on important areas of parts should be carefully considered. Samples should be carefully compared with a specimen of known acceptable quality.

(d) The term "Gage" under "Inspection method" indicates the existence of a Government gage design (see Index of Inspection Equipment Lists, Drawing IEL7101289).

(e) Examination for packaging defects specified in 4.3.2.3.3 shall apply to each link or container; as applicable.

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4.3.2.3.1 *Body, Inner (Drawing D7147027).*

Categories	Defect	AQL	Inspection method	Zone on Drawing
Critical: None defined.				
Major:				
101	Perpendicularity requirement not met, body arm.	0.65	Gage	B–6

4.3.2.3.2 *Link Cartridge, Metallic Belt, 20MM, M12 (Drawings D7147028 and D7548085).* See 4.3.2.3.3 for examination of packaging.

Categories	Defect	AQL	Inspection method	Zone on Drawing
Critical: None defined.				
Major:				
101	Cracks, tears, and malformations.	0.065	Visual	(See 6.5)
102	Missing or improper weld buttons (for welded link only).	0.065	Visual-Manual	Two buttons securely welded, B–6, D7147028
103	Width; end, stress arm (w/o ring).	0.25	Gage	B–5, D7548085
104	Max. clearance between bridge or body arms.	0.25	Gage	C–3, D7548085
105	Min. distance from bridge arms or body arms to vertical center line of link (w/steel plug in link).	0.25	Gage	C–3, D7548085
106	Min. distance from tip of flared end of body stress arm (w/ring) to longitudinal center line of link (w/steel plug in link).	0.25	Gage	B–5, D7548085
107	Grip on cartridge (min. opening).	0.65	Gage	C–5, D7147027
108	Height of bridge tip from bridge base, section BB of assembly drawing.	0.65	Gage	C–3, D7147028
109	Width; tip of stress bridge (for non-welded link only).	0.65	Gage	B–5, D7548085
110	Guide tabs offset height; inner surface max., and outer surface min. (w/ steel plug in link before retention test).	0.65	Gage	D–3, D7147027
111	Width; flared and stress arm (w/ring).	1.0	Gage	C–6, D7548085
Minor:				
201	Missing or defective protective coating.	1.5	Visual	
202	Missing or improper identification.	1.5	Visual	
203	Distance between extreme inside width of body stress arms.	1.5	Gage	B–2, D7147028 B–3, D7548085 (See Dwg. note.)
204	Overall length; guide tabs.	1.5	Gage	B–4, D7147027
205	Overall length of link.	1.5	Gage	B–4, D7147027
206	Inside diameter of connector ring.	1.5	Gage	A7147026
207	Improper clearance; feeder channel guide.	1.5	Gage	B–3 and B–2, D7147027
208	Location: front end of front guide tab (w/steel plug in link).	2.5	Gage	B–4, D7147027
209	Max. overall length; longitudinal offset of guide tabs (w/steel plug in link).	2.5	Gage	B–4, D7147027
210	Min. distance; front surface of ramming guide embossment to rear surface of rear guide tab longitudinal offset.	2.5	Gage	B–4, D7147027
211	Workmanship.	4.0	Visual	

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4.3.2.3.3 *Packaging.* Unless otherwise specified in each listed defect, the packaging requirements are specified on Packaging Data Sheet P7147028 or P7548085, as applicable.

Categories	*Defect*	*AQL*
Critical: None defined.		
Major:		
101	Improper level of packaging and packing (see procurement documents)	1.0
102	Incorrect or illegible marking	1.0
103	Inadequate cleaning and drying	1.5
104	Improper preservation oil application and drainage	1.5
105	Improper containers	1.5
106	Container interior contaminated with dirt, lint, or other foreign matter	1.5
107	Improperly oriented links in unit package	1.5
108	Missing volatile corrosion inhibitor treated sheets between layers of links (Levels A and B only)	1.5
109	Incorrect quantity of links in shipping container	1.5
110	Improper closure of shipping container	1.5
Minor:		
201	Workmanship (see 5.3)	4.0

4.3.3 *Testing.* In addition to the applicable test provisions specified in Specification MIL–W–13855 and other tests required by applicable drawings and specifications, the following tests shall apply. Unless otherwise specified herein, tests shall be performed on samples taken from each inspection lot. Test fixtures and equipment shall be in accordance with Government drawings or, in the absence of drawings, shall be approved by the procuring agency.

4.3.3.1 *Material testing.* Tests required by applicable drawings and specifications shall be performed on materials used in the manufacture of links. The contractor shall furnish the Government representative certified copies of chemical and check analyses and reports of physical tests as may be required for each lot of each material used. The contractor shall also furnish the Government representative certification that the materials used in packaging conform to the applicable specifications.

4.3.3.2 *Accelerated corrosion testing.* Accelerated corrosion testing shall be performed by the contractor in accordance with applicable drawings and specifications.

4.3.3.3 *Hardness testing.* Samples of links taken from each heat-treat lot in accordance with Standard MIL–STD–105, using size code letter G regardless of the quantity in the heat-treat lot, shall be tested by the contractor for the hardness requirements specified on the applicable drawing using the test method specified in 4.4.1. A heat-treat lot shall consist of links heat-treated in a batch-type furnace at one charge or in a continuous-type furnace during an uninterrupted period extending over not more than one shift. Acceptance shall be based on an AQL of 0.40 percent defective.

4.3.3.4 *Weld testing.* Welding shall be tested by the contractor using the test method specified in 4.4.2. Sampling shall be in accordance with Specification MIL–W–45223.

4.3.3.5 *Pull testing.* Samples of links shall be taken from each shift's production in accordance with Standard MIL–STD–105, inspection level S4 and the double sampling plan. Samples shall be tested by the contractor for pull load, using the test method specified in 4.4.3. Acceptance shall be based on an AQL of 0.40 percent defective.

4.3.3.6 *Twist testing.* Four belts of 25 links each taken from each shift's production shall be tested by the contractor for twist requirements using the test method specified in 4.4.4. Failure of any belt to meet the requirements shall cause rejection of the represented shift's production.

4.3.3.7 *Cartridge retention testing.* Samples of links shall be taken from each shift's production in accordance with Standard MIL–STD–105, inspection level S4 and the

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double sampling plan. Samples shall be assembled and tested by the contractor for cartridge retention using the test method specified in 4.4.5. Acceptance shall be based on an AQL of 0.40 percent defective.

4.3.3.8 *Free hinging testing.* Four belts of 25 links each taken at random from each shift's production shall be tested by the contractor for free hinging using the test method specified in 4.4.6. If two or more links fail to meet the requirements, the represented shift's production shall be rejected. If one link fails to meet the requirements, a retest shall be conducted using eight additional belts. Failure of any link in the retest shall be cause for rejection of the shift's production. When 10 consecutive "shift's production" samples have met the above acceptance criteria, the sample size for the free hinging test may be selected on an "inspection lot" basis. Should any subsequent "inspection lot" sample indicate a defective link, the contractor will revert to the sample size and criteria specified for a "shift's production." The contractor shall re-qualify 10 consecutive "shift's production" samples prior to resorting to selection of samples on an inspection lot basis.

4.3.3.9 *Functioning testing.* A sample of 200 links taken from the initial production sample and a sample of 100 links from each monthly production sample (see 3.1) shall be tested for functioning by the testing agency using the test method specified in 4.4.7. Links failing to function properly shall be inspected as necessary to determine compliance with drawing requirements.

4.3.3.10 *Determination of cleanliness testing.* Links shall be tested by the contractor for cleanliness using the test method specified in 4.4.8. Sampling shall be in accordance with Specification MIL–P–116.

4.3.4 *Inspection equipment.* Use of the contractor's or Government furnished inspection equipment, when desired by the Government representative, shall be permitted without charge.

4.4 Test methods.

4.4.1 *Hardness test.* Each link of the sample shall be tested for compliance with drawing requirements for hardness using the test method 243 of Federal Test Method Std. No. 151. All hardness testing shall be conducted prior to phosphating.

4.4.1.1 *Welded links.* Welded links shall be tested for hardness requirement (see 3.2.1) on top of the stress bridge outside the welded area. Samples shall be taken from link bridges removed during the weld test.

4.4.1.2 *Nonwelded links.* Nonwelded links shall be tested for hardness requirement (see 3.2.1) after heat treatment by conducting separate tests on the bridge and body.

4.4.2 *Weld test.* Welding shall be tested for compliance with the applicable drawing (see 3.2.2) and in accordance with Specification MIL–W–45223.

4.4.3 *Pull test.* Each link of the sample shall be tested for pull requirement (see 3.2.3) using the inspection equipment specified on Drawing F7190358. The method of use as described on the fixture drawing shall be followed. The links shall be inserted in the fixture and subjected to the required load as specified on the applicable drawing.

4.4.4 *Twist test.* The links shall be tested for twist requirement (see 3.2.4) by assembling the links into belts using Government standard 20MM dummy cartridges. Each belt shall be laid out horizontally to its full length, with the open side of the links down, and one end of the belt shall be flipped sharply by hand 180 degrees to its other side so that the twisting action will progressively move along the belt to the free end. The test shall be repeated with the belt positioned with the open side of the links up.

4.4.5 *Cartridge retention test.* Each link of the sample shall be assembled and tested for cartridge retention requirement (see 3.2.5) using the inspection equipment and method specified on Drawing F7190403. Partial extraction shall be classed as failure (see 6.5.6).

4.4.6 *Free hinging test.* The links shall be tested for free hinging requirement (see 3.2.6) by assembling the links into belts

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using Government standard 20MM dummy cartridges as specified on applicable drawing. Each belt shall be laid out horizontally to its full length, with the open side of the links up, and one end shall be drawn back upon the remainder of the belt until the belt is completely reversed (open side of links down). One end of the reversed belt shall then be drawn back upon the remainder of the belt until the belt is returned to its original (open side of links up) position.

4.4.7 *Functioning test.* The sample links shall be tested for functioning requirement (see 3.2.7) by assembling the links into belts using Government standard 20MM cartridges and firing the belts in an M39A2 automatic gun. Firing shall be in bursts of 50 rounds each.

4.4.8 *Determination of cleanliness test.* The sample links shall be subjected to the determination of cleanliness test of Specification MIL–P–116.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. On each contract a pilot pack consisting of a complete and packed shipping container representing initial production shall be forwarded in accordance with 3.1.

5.2 Preparation for delivery shall be in accordance with Packaging Data Sheet P7147028 or P7548085, as applicable, using the levels of preservation, packaging, and packing specified in the contract and shall conform to the workmanship requirements specified herein.

5.3 Workmanship. Adequate controls shall be utilized to check for contamination of cleaning solvents and preservative.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) List of drawings and specifications pertinent to the link showing applicable revision dates.
(c) Index of Inspection Equipment Lists pertinent to the link showing applicable revision dates.
(d) Selection of applicable levels of preservation, packaging, and packing (see 5.2 and 6.4).
(e) Place of final inspection and acceptance.
(f) Responsibility for acquisition, maintenance, and disposition of acceptance inspection equipment, when required.
(g) Responsibility for furnishing dummy cartridges.

6.2 When action by a testing agency is required, work programing will be effected with the testing agency at the earliest practicable date.

6.3 Normally, the lot number will consist of three parts—a manufacturer's identification symbol, interfix number, and a serial number, each separated by hyphens. The identification symbol consists of letters assigned so as to indicate the identity of the manufacturer. The interfix number is used to identify lots with respect to variations in the design or manufacturing process. The serial number identifies the lot according to its sequence of production under the conditions of manufacture represented by the interfix.

6.4 When links are prepared for domestic shipment from manufacturers to ammunition loading plants for immediate use, level C preservation, packaging, and packing, as specified on Packaging Data Sheet P7147028 or P7548085, as applicable, should be used.

6.5 Definitions of terms.

6.5.1 *Malformed links.* Links whose shape or dimensions or both have been mutilated or altered by processing to the extent that the links are unsuitable for their intended use.

6.5.2 *Cracks.* See figures 1, 2, and 3.

6.5.2.1 *Cracks in principal areas.* Any visible fracture of the link extending through the stock thickness in depth or a distance in length equal to or greater than the material stock thickness.

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6.5.2.2 *Cracks in nonprincipal areas.* Any visible fracture of the link extending through the stock thickness in depth and equal to or greater than the stock thickness in length; or any visible fracture less than stock thickness in depth and greater than two times stock thickness in length.

6.5.3 *Tears.*

6.5.3.1 *Tears in principal areas.* Any visible fracture of the link with ragged or irregular edges extending through the stock thickness in depth or a distance in length equal to or greater than the material stock thickness.

6.5.3.2 *Tears in nonprincipal areas.* Any visible fracture of the link with ragged or irregular edges extending through the stock thickness in depth and equal to or greater than the stock thickness in length; or any visible fracture less than stock thickness in depth and greater than two times stock thickness in length.

6.5.4 *Burrs.* Any rough edges on any part of the link which, in the opinion of the Government representative, would produce a definite removal of metal from a cartridge case or which could injure personnel in handling.

6.5.5 *Principal areas.* The body arms (except for the edges facing inward between the flared ears), stress bridge arms, and the connecting ring.

6.5.6 *Partial extraction.* When only one end of the mandrel or round becomes free.

Custodians:
Army—WC
Air Force—OOA

Interested activities:

Reviewers:
Army—MU WC
Air Force—70

Preparing activity:
Army—WC

Project No. 1305–0152

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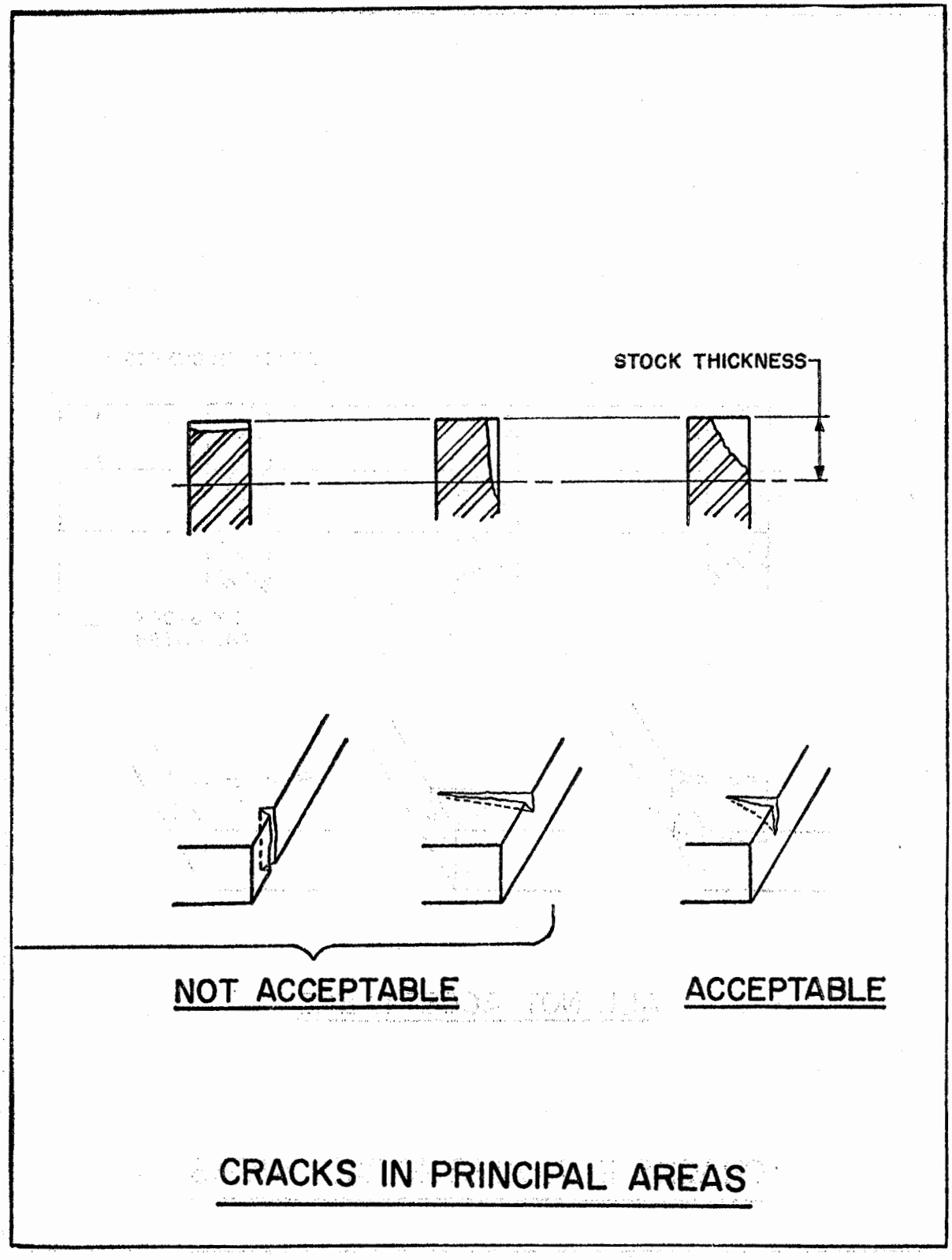


FIGURE 1.

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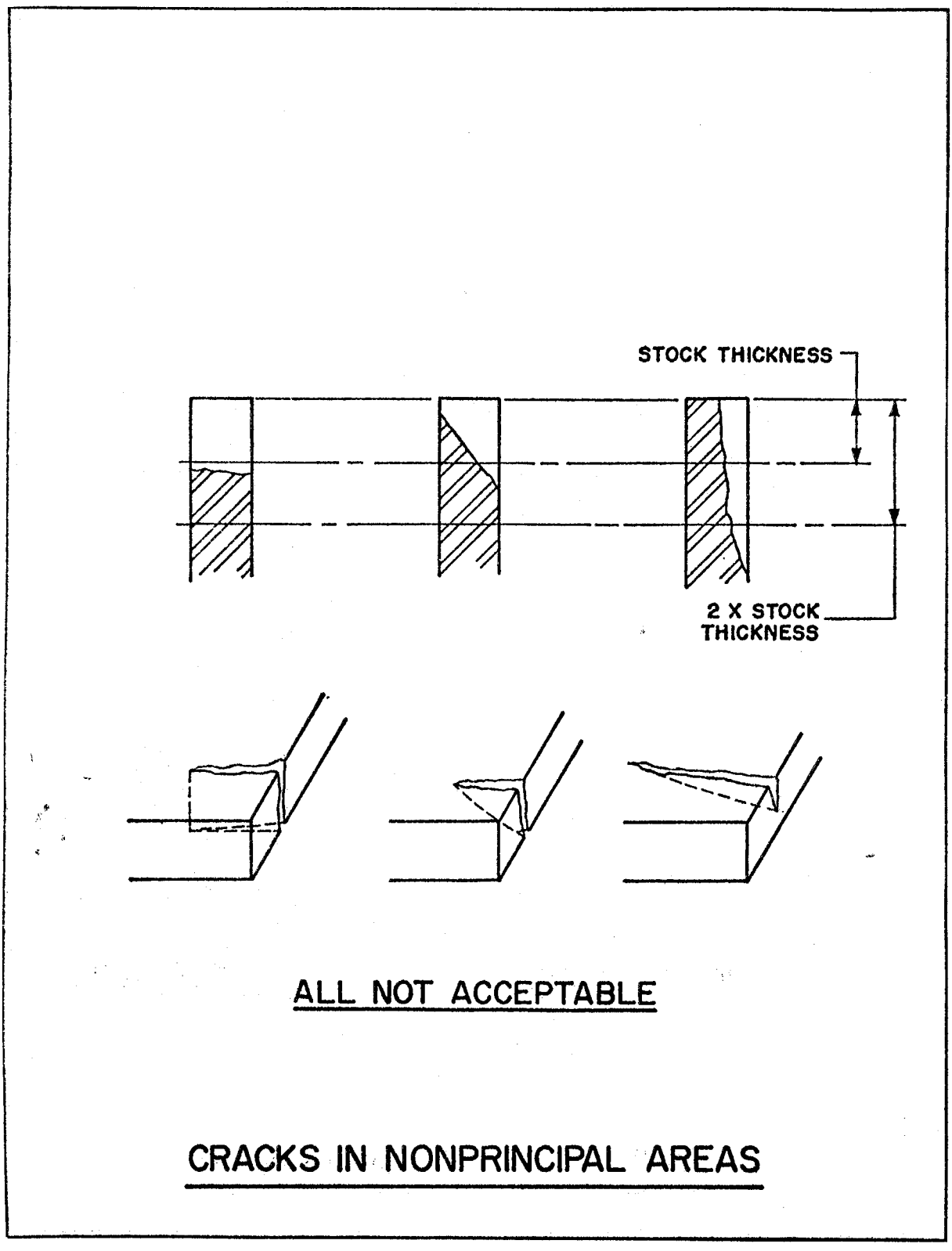


FIGURE 2.

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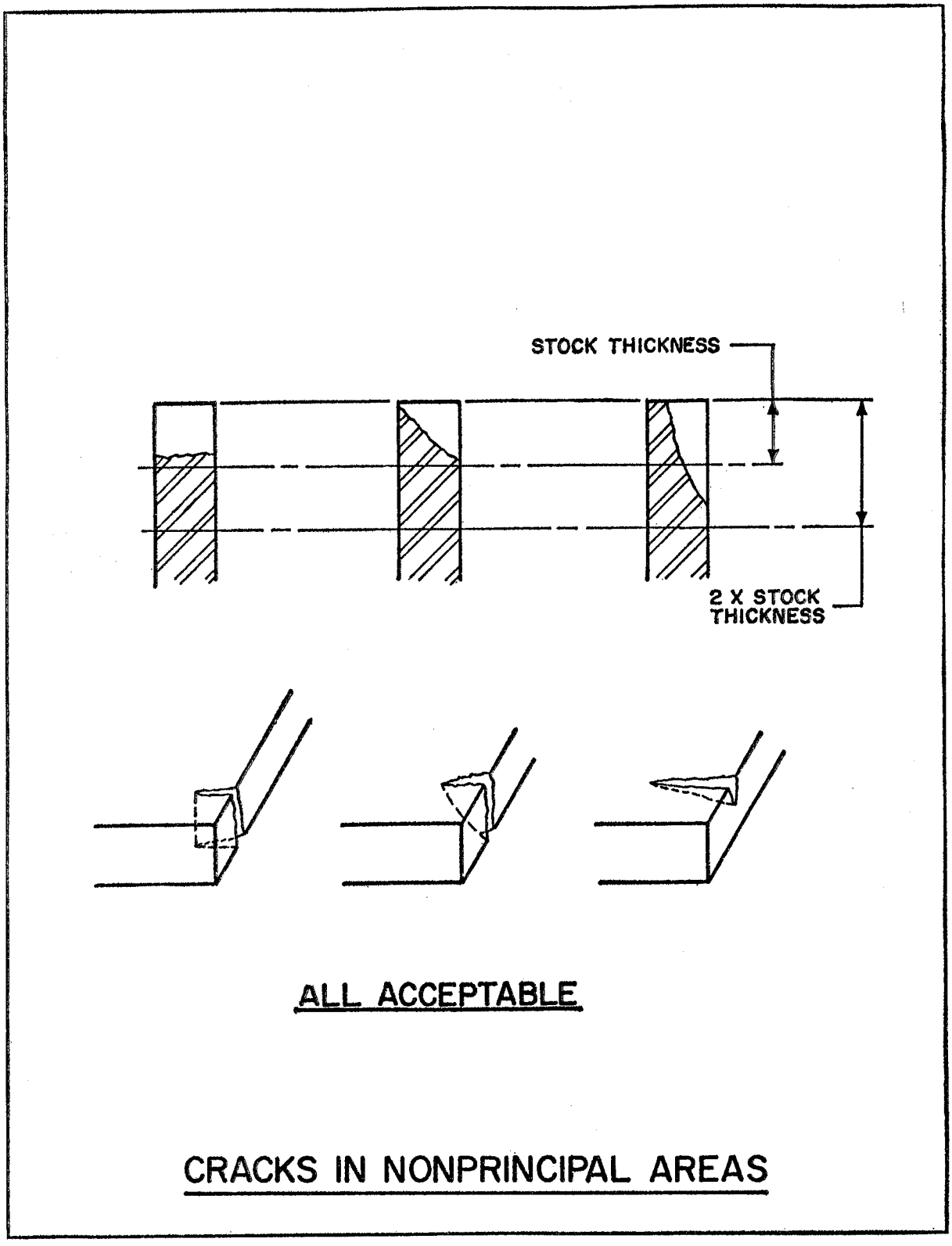


FIGURE 3.

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